Superposition and interference

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add signed displacements and distinguish interference from lost energy.

SUCCESS CRITERIA - I can...

1. Add wave displacements at the same place and time.
2. Recognize constructive and destructive interference.
3. Explain that pulses continue after crossing in a linear medium.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

An upward rope pulse is 3 cm high before reaching an ideal free end. Is the reflected displacement upright or inverted? Sketch it.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $y_{resultant} = y_1 + y_2$.
Add signed displacements at the same place and time. Local cancellation does not destroy all wave energy.

Worked reasoning: At one point, two pulses have displacements +3 cm and -1 cm. Superposition gives +2 cm there. Add signed displacements, not their absolute values.

Your turn: Draw the two component displacements and the resultant at that point.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two upward pulses overlap. Sketch the combined displacement. Then sketch equal upward/downward pulses overlapping.



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

Two rope pulses overlap. At point A their displacements are +3 cm and -1 cm; at point B they are +2 cm and -2 cm. Add signed displacements at each point. Draw the two contributions and each resultant.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

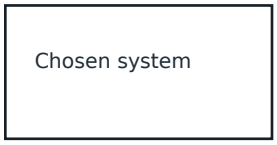
Two rope pulses momentarily cancel. A student says the pulses are gone forever. What happens after they pass in an ideal linear medium?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A system receives 60 J heat and does 40 J work on its surroundings. Find ΔU with $\Delta U = Q - W_{by}$.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

07 Exit check

Add displacements -4 mm and +7 mm. Name whether they reinforce or partly cancel.

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Construct three different pairs of displacements with resultant zero. Can the waves still move after that instant?

A

B

Sum

Add signed displacements; label units.

Marzano check - circle: 1 2 3